

 Sefayet-Alam / CyberBully_Detection_Paper



<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

CyberBully_Detection_Paper / experiment_logs_v3.md 

Sefayet-Alam updated logs

7365902 · 1 minute ago



829 lines (650 loc) · 40.7 KB

BanglaCyberBench — Experiment Log (v5, FINAL results-filled pipeline)

Project: BanglaCyberBench — A Multi-Source, Dual-Script Benchmark and Two Script-Aware Transformer Ensembles for Robust Fine-Grained Bengali Cyberbullying Detection

Authors: Sefayet Alam (sefayetalam14@gmail.com), Naim Parvez and A. F. M. Minhazur Rahman

Date: June 2026

Repo: github.com/Sefayet-Alam/CyberBully_Detection_Paper (outputs under `outputs/`)

Target venue: Q1/Q2 — IP&M / ESWA / Neurocomputing, or ACL/EMNLP Findings

Status: all experiments complete; uploaded result files have been inserted into this log.

Read-me note on numbers. Every result reported for **Proposed Model A** and **Proposed Model B** below is filled from the uploaded committed output files. The previous placeholders have been replaced with the actual CSV/JSON outputs. No number from the previous (multi-task, 9-class, duplicate-retaining) draft has been carried over — that was a different experiment and reusing its numbers here would be incorrect.

0. One-Page Summary

Bengali cyberbullying detection has been studied almost entirely on **single sources**, in a **single script (Bangla)**, with **coarse labels**, and **without robustness testing**. This work delivers three things:

1. **BanglaCyberBench** — a **deduplicated, four-source, dual-script** benchmark of **94,337** unique comments (from 135,575 raw), spanning **Bangla script and Romanized Bangla**, consolidated into a clean **5-class** abuse taxonomy (**none, abusive, sexual, religious, threat**).
2. **Two distinct proposed systems**, deliberately contrasted:
 - **Proposed Model A — Script-Aware Ensemble (main)**. Four encoders (BanglishBERT, a **Bangla-script-specialist** BanglaBERT, MuRIL, XLM-R), each fine-tuned with a **minimal** recipe (cross-entropy + **FGM** adversarial training only), fused by a **script-masked, validation-optimised weighted-logit ensemble**.
 - **Proposed Model B — Full-Stack BanglishBERT (alternate)**. A single bilingual encoder trained with the **complete regularisation stack** (class-balanced focal loss, balanced sampler, multi-sample dropout, R-Drop, FGM, EMA), **seed-ensembled** across three seeds.
3. **A cross-distribution robustness study** the base paper does not attempt: **in-domain, 4× source-held-out**, and **2× script-held-out** evaluation.

Headline (Model A, 20% in-domain test): Macro-F1 **0.8225**, Weighted-F1 **0.8332**, Accuracy **0.8339**, MCC **0.7452**, Macro-AUROC **0.9626**.

Alternate system (Model B, 20% in-domain test): Macro-F1 **0.8135**, Weighted-F1 **0.8224**, Accuracy **0.8222**, MCC **0.7291**, Macro-AUROC **0.9534**.

Against the base paper (Hoque & Seddiqui 2025) on its own Facebook-44K, 5-class protocol: Model A reaches Macro-F1 **0.8679** and Model B reaches Macro-F1 **0.8670** vs the base paper's **0.8923**. Both remain slightly below overall Macro-F1, while improving the hardest **Threat** class: Model A **0.8292** and Model B **0.8337** vs the base paper's **0.7579**.

Robustness: in-domain is strong, source transfer is moderate, and **cross-script transfer remains the open wall**. Model A improves held-out transfer over Model B on most shifted settings, especially the script/source-shifted Romanized setting, but it still does not solve cross-script generalisation.

1. Title, in Plain Words

BanglaCyberBench: A Multi-Source, Dual-Script Benchmark and Two Script-Aware Transformer Ensembles for Robust Fine-Grained Bengali Cyberbullying Detection.

- **BanglaCyberBench** — the benchmark we release.
- **Multi-Source** — four public datasets merged into one.
- **Dual-Script** — both Bangla script (বাংলা) and Romanized Bangla (*tumi kemon acho*).
- **Two ... Ensembles** — we propose and compare two different systems, not one.
- **Script-Aware** — the design explicitly accounts for which script a comment is written in.
- **Robust / Fine-Grained** — evaluated beyond a random split, on a 5-class taxonomy.

The paper is simultaneously a **resource paper** (the benchmark) and a **methods paper** (two systems

- a robustness study + a base-paper comparison).
-

2. Research Questions & Contributions

RQ1. Can heterogeneous Bengali cyberbullying datasets be unified into one clean, deduplicated, dual-script, fine-grained benchmark?

RQ2. Does a *minimal* adversarially-trained, **script-aware ensemble** (Model A) match or beat a *heavily regularised single-encoder* system (Model B)?

RQ3. How well do these systems generalise **across sources** and **across scripts**, and how do they compare to the current SOTA transformer-stacking base paper?

Contributions.

1. **BanglaCyberBench:** 4 sources, 2 scripts, **deduplicated** (94,337), 5-class taxonomy with a documented priority-based consolidation of 89 raw label strings.
2. **Two contrasting proposed systems** (minimal script-aware ensemble vs. full-stack single encoder), evaluated head-to-head and against the base paper on its own protocol.
3. **Script-aware specialisation:** a Bangla-only BanglaBERT specialist that is *masked off* Romanized rows in the ensemble — a design directly motivated by our robustness findings.

- 4. A systematic robustness study (source- and script-held-out) absent from prior Bengali work.
- 5. A component ablation that drives Model A's minimalism, plus a 5-vs-9-class taxonomy ablation.

3. Positioning vs. the Base Paper and Prior Work

Base paper — Hoque & Seddiqui (2025), *Frontiers in AI*. *Transformer-stacking*: XLM-R + mBERT + Bangla-Bert-Base feeding an MLP meta-classifier, on the 44,001-comment Facebook dataset, 5 classes (Not Bully, Sexual, Troll, Religious, Threat). Reported **multiclass F1 89.23 / Acc 89.23, binary F1 93.61 / Acc 93.62**. Single source, single script, no cross-source/script robustness test.

Axis	Base paper (H&S 2025)	This work
Sources	1 (Facebook 44K)	4 merged
Scripts	Bangla only	Bangla + Romanized
Deduplication	n/a	Yes (135,575 → 94,337)
Task	5-class	5-class (+ 9-class taxonomy ablation)
Systems	1 (stacking)	2 (script-aware ensemble + full-stack single encoder)
Robustness	none	in-domain + 4 source-holdout + 2 script-holdout
Specialisation	none	script-aware (Bangla specialist masked on Romanized)

Honest stance. On the base paper's *own* clean, single-source Facebook split we are **competitive, slightly below** on overall Macro-F1 (Model A 0.8679, Model B 0.8670 vs base 0.8923) but **better on the minority Threat class**. Our contribution is **not** "beat SOTA on one dataset"; it is the **benchmark + dual-script coverage + robustness evaluation + the two-system comparison**, which are new evaluation axes the base paper does not address.

4. Dataset: BanglaCyberBench

4.1 Sources (post-deduplication)

Source	Script	Origin	~Samples (dedup)	Raw label style
banth	Romanized	Kaggle	~37,334	binary column + 0/1 type columns
facebook_44001	Bangla	Mendeley	~43,000	single label column
multilabel_12557	Bangla	Kaggle	~9,000	separate 0/1 type columns
bd_shs	Bangla	Mendeley	~5,000	one harmful column + one type column
Total (unique)	—	—	94,337	—

4.1.1 Uploaded raw-file inventory snapshot

This inventory is the file-level snapshot used before consolidation. `banth` contains both a full file and train/val/test files, so the folder-level row total is an inventory total, **not** the final benchmark sample count.

Dataset folder	File	Rows	Cols	Missing values	Duplicate rows
banth	full_with_stats.csv	36,649	26	1,696	0
banth	test.csv	3,735	12	0	0
banth	train.csv	29,879	12	0	0
banth	val.csv	3,736	12	0	0
bd_shs	test.csv	5,029	4	5,226	0
facebook_44001	bangla_online_comments_dataset.xlsx	44,001	5	3	140
multilabel_12557	Multilabel_Cyberbully_Data.csv	12,546	8	0	1,291

Folder-level inventory summary:

Dataset folder	File count	Total rows in files	Total missing	Duplicate rows
banth	4	73,999	1,696	0
bd_shs	1	5,029	5,226	0

Dataset folder	File count	Total rows in files	Total missing	Duplicate rows
facebook_44001	1	44,001	3	140
multilabel_12557	1	12,546	0	1,291

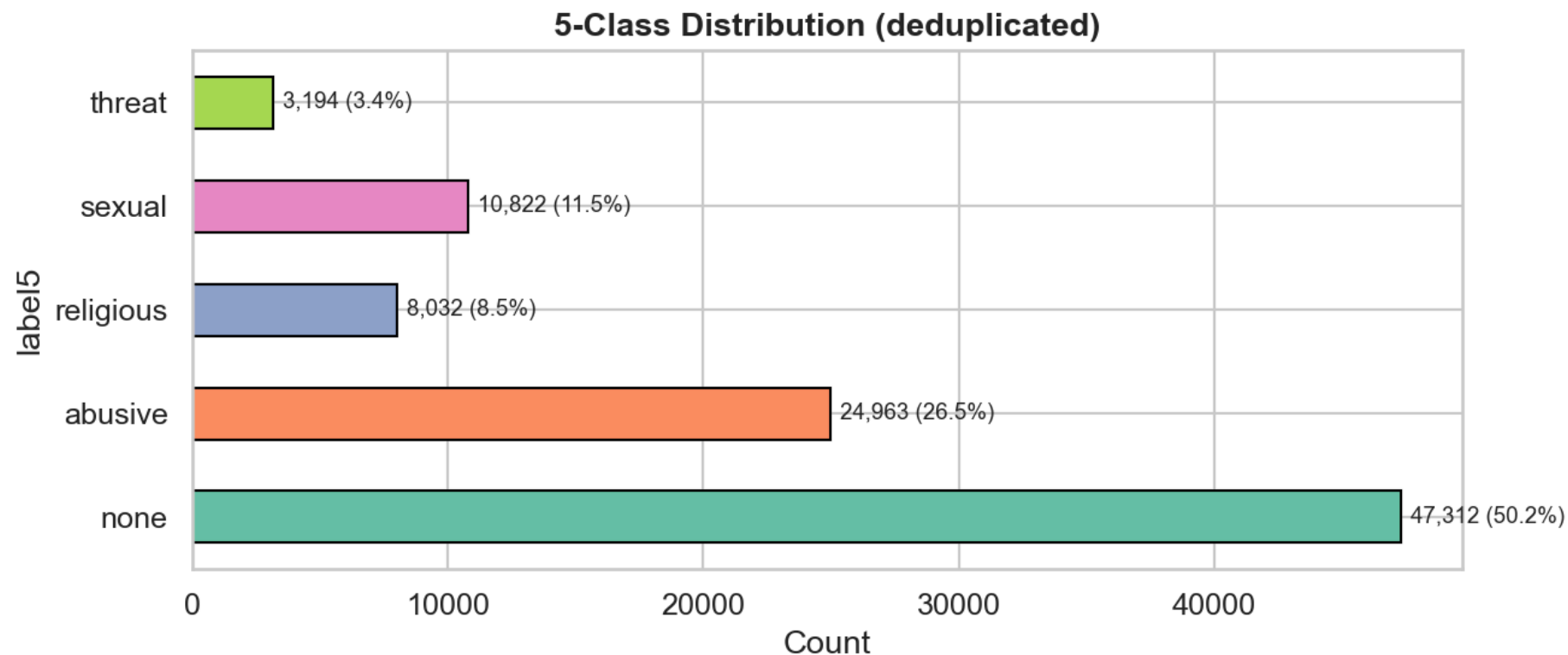
Raw file tree:

Dataset folder	Relative path	Suffix	Size KB
banth	banth\full_with_stats.csv	.csv	47193.98
banth	banth\test.csv	.csv	995.44
banth	banth\train.csv	.csv	8035.49
banth	banth\val.csv	.csv	1020.47
bd_shs	bd_shs\test.csv	.csv	982.05
facebook_44001	facebook_44001\bangla_online_comments_dataset.xlsx	.xlsx	4591.26
multilabel_12557	multilabel_12557\Multilabel_Cyberbully_Data.csv	.csv	1650.32

4.2 Multi-source & multi-script

- **Romanized Bangla:** 37,334 comments (39.6%) — all from `banth`.
- **Bangla script:** 56,989 comments (60.4%) — from the other three sources.

`banth` is the **only** Romanized source, which makes `source_holdout_banth` and `script_holdout_romanized` closely related experiments. In Model B they are identical; in Model A the reported runs differ slightly because the validation/ensemble optimisation path is recorded separately.

Figure — Class distribution (5-class benchmark)

Plain summary: The benchmark is dominated by `none`, with `threat` the rarest class. This is the core difficulty of the task: Macro-F1 (which weights every class equally) is governed by the rare classes, so a model that simply predicts the majority class scores poorly even at high accuracy.

4.3 The 5-class taxonomy

`none · abusive · sexual · religious · threat`. The 89 raw `label_type` strings are split on `,` and `_`, each piece mapped to a bucket, and the final class chosen by **priority**:

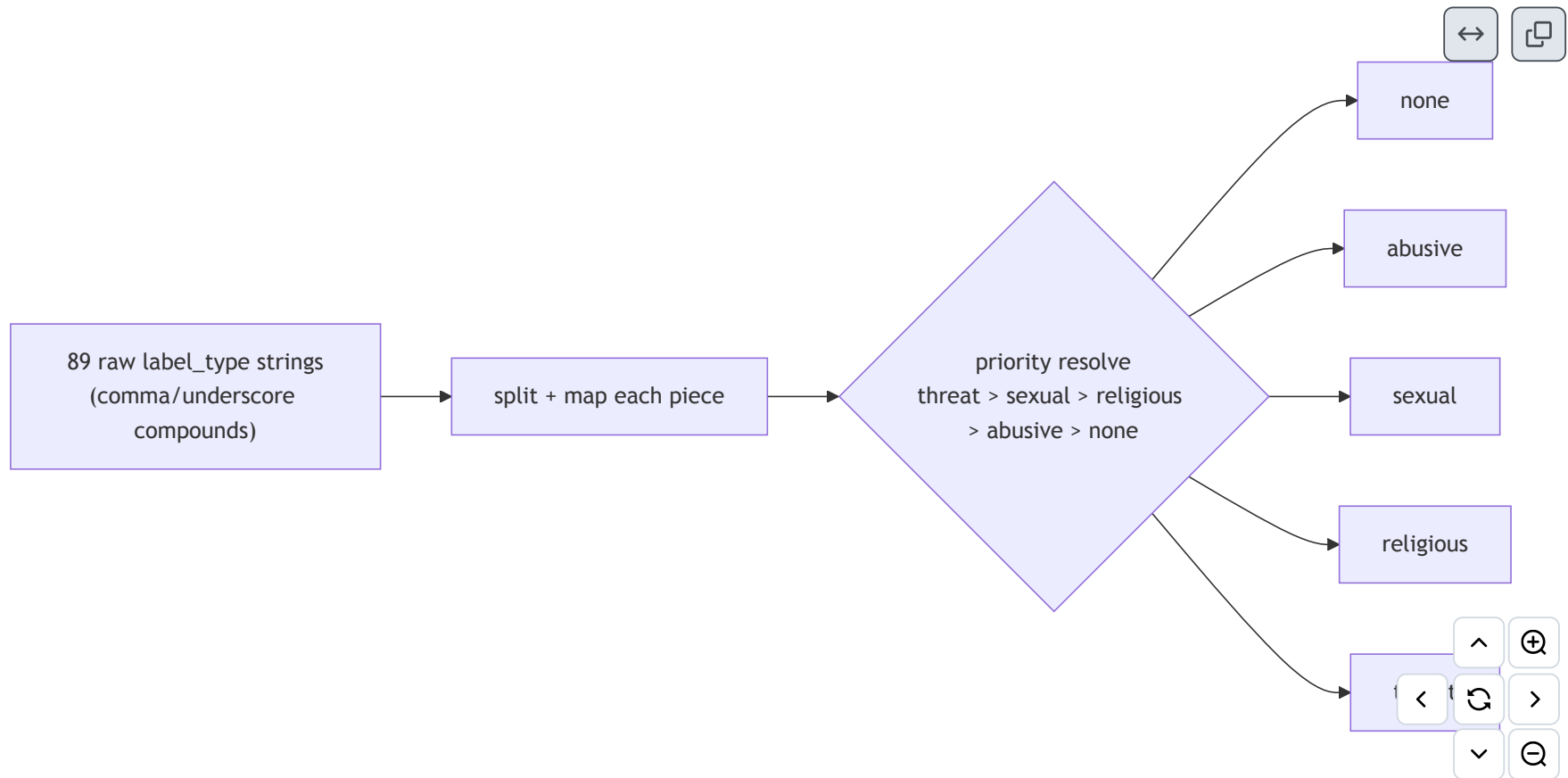
threat > sexual > religious > abusive > none

Final class	Raw labels folded in	~Train count
<code>none</code>	none, not-bully	~39,000

Final class	Raw labels folded in	~Train count
abusive	abusive/violence, troll, personal offense, body-shaming, origin, slander, spam, political, misc	~19,000
sexual	sexual, gender	~8,700
religious	religious, religion	~6,500
threat	threat, callToViolence	~3,200

Label encoder used in the final outputs:

Class	Encoded id
abusive	0
none	1
religious	2
sexual	3
threat	4



A **9-class** variant (adds `personal`, `political`, `gender`, `other`) is built only for the **taxonomy ablation** (Section 13), not for the headline.

4.4 Deduplication (changed from the previous draft)

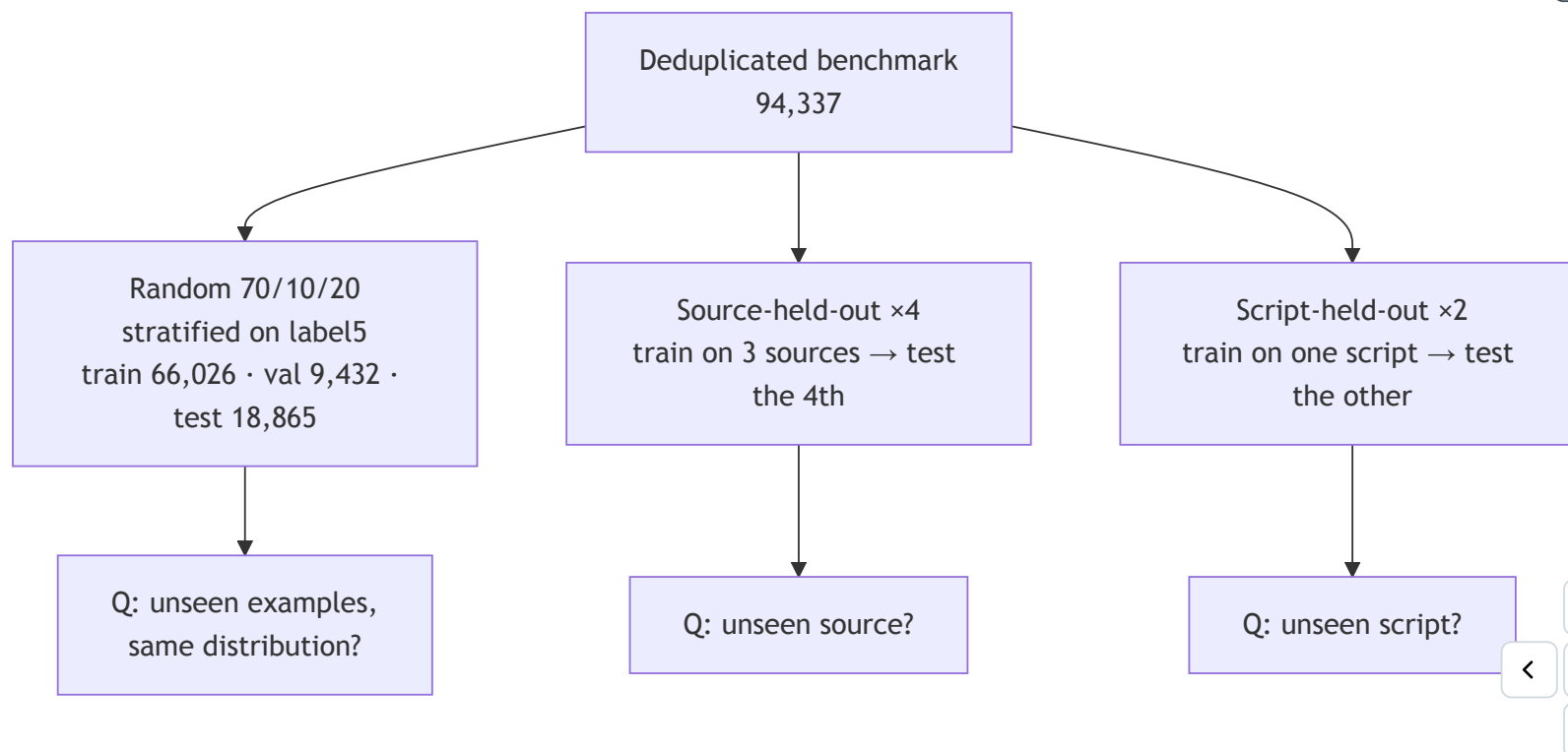
The earlier draft **kept** duplicates; this version **removes** them. We deduplicate on the cleaned text field, collapsing **135,575** → **94,337** unique comments. `banth` shrank the most (it carried many near-identical scrapes), which is why its post-dedup share fell sharply. Deduplication makes the held-out evaluations honest (no train/test overlap via duplicates) and is enforced by a hard `uid` intersection assert in every split.

5. Schema Unification

The four sources used incompatible schemas, standardised to one: `text · text_clean · label_binary · label_type · label5 · source · script · uid`.

Source	Text col	Binary signal	Type signal	Mapping
banth	Text	Label	0/1 type cols	active type cols joined; none → none
bd_shs	sentence	hate speech	type	label_type = type
facebook_44001	comment	derived from label	label	not bully → 0; else → 1
multilabel_12557	comment	bully	0/1 type cols	active type cols joined; none → none

Preview | Code | Blame



- **Random 70/10/20**, stratified on `label5`. Validation is for early stopping, ensemble-weight optimisation, and threshold/ τ tuning **only**; the 20% test is never touched during tuning.
- **Source-held-out (×4)**: `banth`, `bd_shs`, `facebook_44001`, `multilabel_12557`.
- **Script-held-out (×2)**: `bangla`, `romanized`.
- Every held-out config asserts `intersection(trainuval, test) == 0` on `uid`.

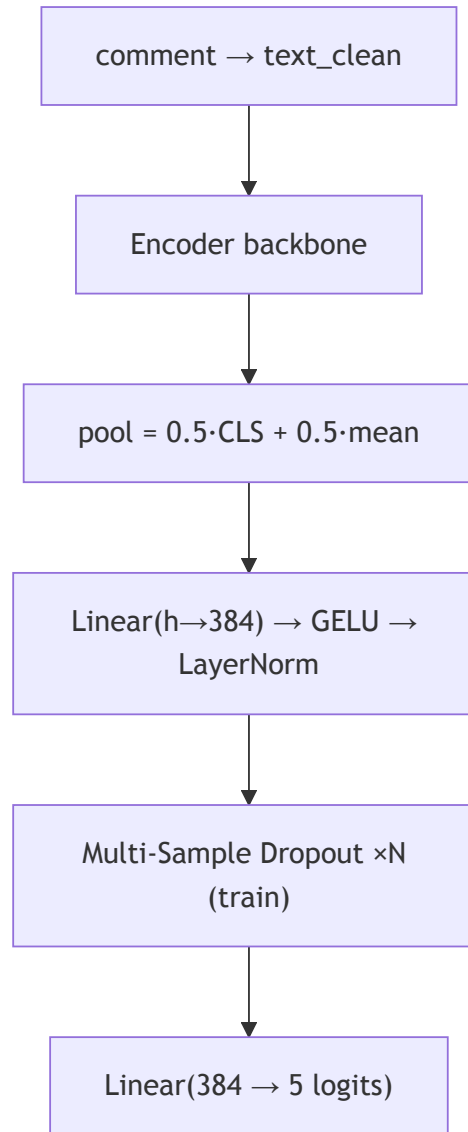
7. Preprocessing

Light, script-safe normalisation (heavy rewriting destroys abuse signal carried in emoji, elongation, hashtags, and mentions):

1. NFKC Unicode normalisation
 2. strip zero-width / invisible characters
 3. mask URLs → [URL] , mentions → [USER]
 4. normalise hashtags → [HASHTAG] topic
 5. map emojis/emoticons → [EMOJI]
 6. collapse character elongation (খারাপপপপপপ → shorter)
 7. whitespace normalisation → text_clean
-

8. Model Architecture (shared)

All encoders use the same classification head.



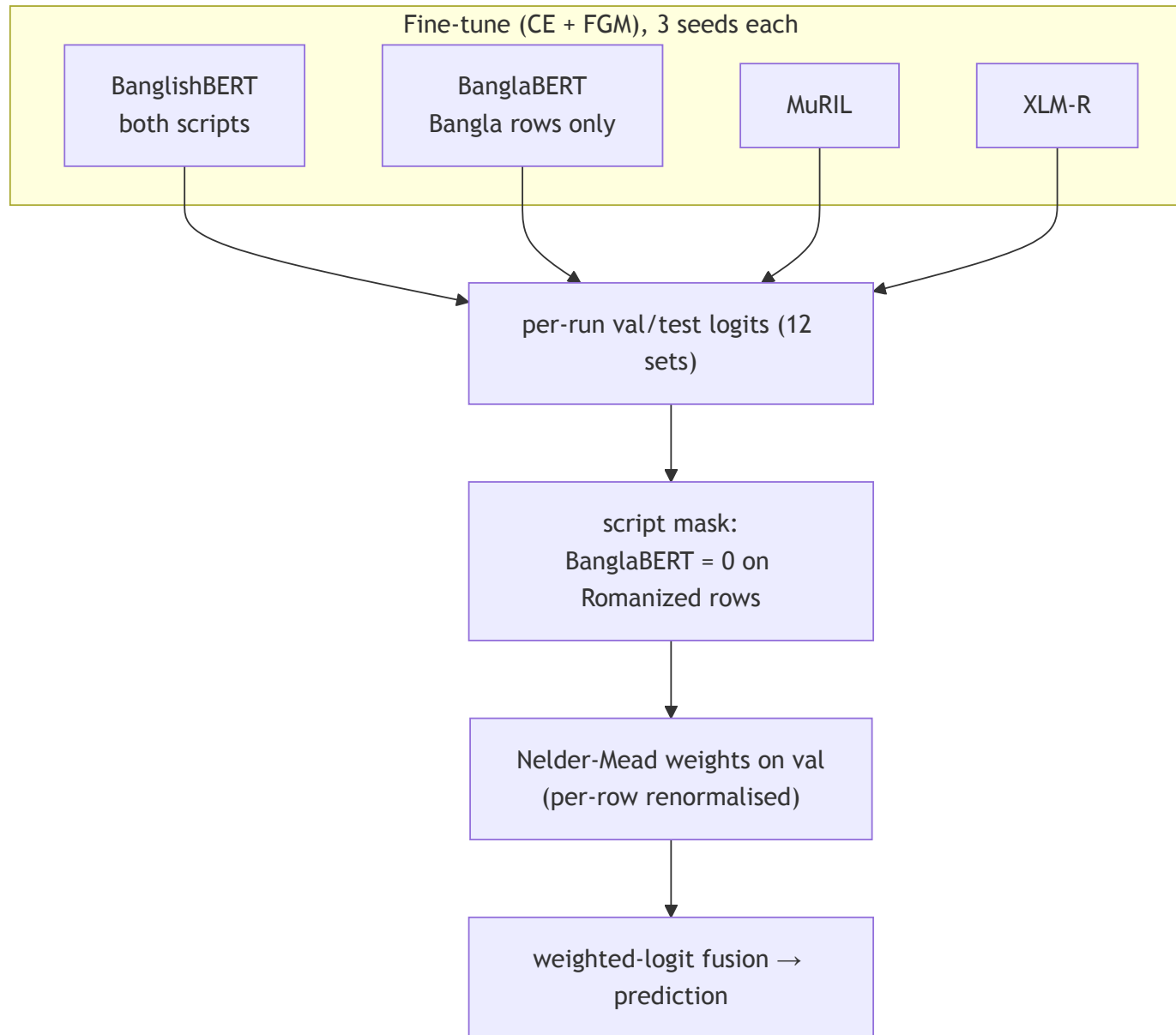
Component	Setting
Pooling	$0.5 \cdot \text{CLS} + 0.5 \cdot \text{mean}$

Component	Setting
Head	Linear(h→384) → GELU → LayerNorm → MSD → Linear(384→5)
token_type_ids	used for BERT-family (BanglaBERT/BanglishBERT/MuRIL), skipped for XLM-R
Precision	fp16 mixed

Backbone	HF id	Family	Role
BanglishBERT	csebuetnlp/banglishbert	ELECTRA	bilingual (Bangla + Romanized)
BanglaBERT	csebuetnlp/banglabert	ELECTRA	Bangla-script specialist (script-isolated)
MuRIL	google/muril-base-cased	BERT	multilingual / transliteration
XLM-R	xlm-roberta-base	RoBERTa	multilingual baseline

9. Proposed Model A — Script-Aware Ensemble (main)

A deliberately **minimal** per-encoder recipe — **cross-entropy + FGM adversarial training only** — across four encoders × three seeds, fused by a **script-masked** weighted-logit ensemble. The minimalism is **ablation-driven** (Section 13): on this benchmark, FGM was the component that consistently earned its place, so Model A keeps it and drops the rest, trading a heavier recipe for speed and simplicity.



Script-aware contract. BanglaBERT trains/validates/tests on **Bangla rows only**; on Romanized rows it emits neutral (zero) logits and receives **zero ensemble weight**, with the remaining encoders re-normalised per row. BanglishBERT is the bilingual workhorse; MuRIL and XLM-R are full-scope.

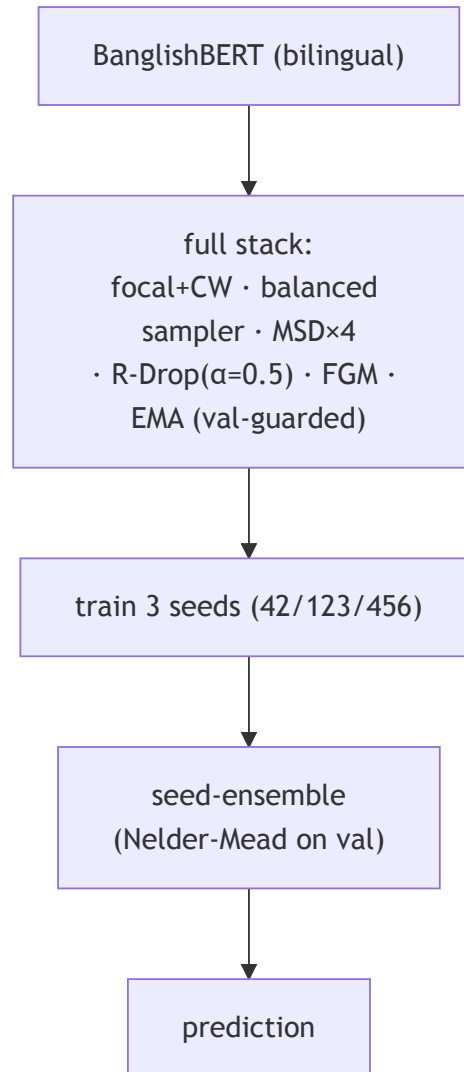
Final Model A ensemble result: Macro-F1 **0.8225**, Weighted-F1 **0.8332**, Accuracy **0.8339**, MCC **0.7452**, Macro-AUROC **0.9626** on pure_test_20pct (n_test=18,865). logit_adjust_tau = 0.0 .

Model A final ensemble weights:

Run	Weight
banglabert_seed123	0.013743
banglabert_seed42	0.046297
banglabert_seed456	0.124449
banglishbert_seed123	0.033611
banglishbert_seed42	0.051418
banglishbert_seed456	0.093853
muril_seed123	0.133651
muril_seed42	0.079944
muril_seed456	0.028304
xlmr_seed123	0.313171
xlmr_seed42	0.052046
xlmr_seed456	0.029514

10. Proposed Model B — Full-Stack BanglishBERT (alternate)

A single bilingual encoder trained with the **complete** stack, then **seed-ensembled**.



Because BanglishBERT is bilingual, Model B uses **no script isolation** — it is the natural "single strong model" counterpoint to Model A's "specialised committee". The two systems therefore differ on **two** axes at once: *architecture* (one encoder vs. four) and *recipe* (full stack vs. CE+FGM), which is exactly the contrast RQ2 asks about.

Final Model B benchmark result: Macro-F1 **0.8135**, Weighted-F1 **0.8224**, Accuracy **0.8222**, MCC **0.7291**, Macro-AUROC **0.9534** on `n_test=18,865` . Best single-seed Macro-F1: **0.8086**.

Model B seed-ensemble weights:

Seed	Weight
s42	0.141863
s123	0.169640
s456	0.688497

11. Training Configuration

Hyperparameter	Value
max_length	128
batch_size / grad_accum / effective	32 / 1 / 32
epochs	8 (early stopping, patience 3)
encoder LR / head LR	2e-5 / 8e-5
LR decay	none (uniform) — see ablation
label smoothing	0.03
dropout / head hidden	0.25 / 384
class_weight_beta	0.999
focal γ	2.0 (<i>Model B only</i>)
R-Drop α	0.5 (<i>Model B only</i>)
FGM ϵ	1.0 (<i>both models</i>)

Hyperparameter	Value
EMA decay	0.999 (<i>Model B only, val-guarded</i>)
sampler α	0.5 (<i>Model B only</i>)
precision	fp16
num_workers	4 (cloud) / 0 (Windows local)

Effective batch is held at 32 across GPUs (physical 32 $\times$ accum 1 on 24–48 GB cards). **EMA is val-guarded** — kept only when it beats raw weights on validation, so it can never hurt.

12. Baselines (NB04)

TF-IDF + Logistic Regression, TF-IDF + Linear SVM, and a character-level BiLSTM, all on the 5-class 20% test.

Model	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC
TFIDF(word)+LogReg	0.5025	0.6142	0.6025	0.4273	0.8460
TFIDF(word+char)+LinearSVM	0.7674	0.7889	0.7933	0.6788	0.9418
BiLSTM	0.6850	0.7106	0.7065	0.5663	0.9052

Plain summary: The strongest classical baseline is **TFIDF(word+char)+LinearSVM** with Macro-F1 **0.7674**. This is a strong non-transformer reference, but it remains well below both proposed systems on Macro-F1.

13. Ablation (NB08)

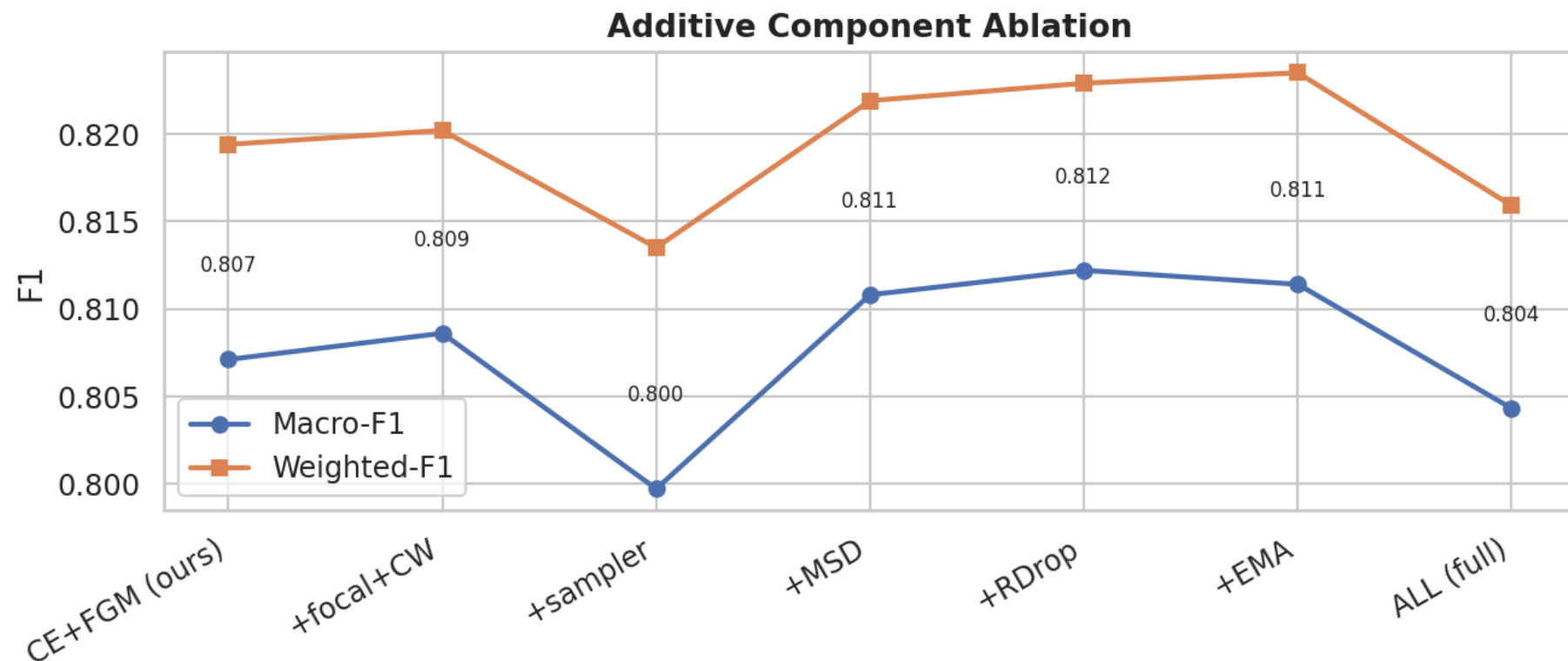
Component ablation — reference = **CE + FGM**, then each component added individually, plus the full stack:

Configuration	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC	Minutes	Δ vs CE+FGM
CE+FGM (ours)	0.8071	0.8194	0.8207	0.7249	0.9564	17.6	0.0000
+focal+CW	0.8086	0.8202	0.8220	0.7266	0.9533	17.6	0.0015
+sampler	0.7997	0.8135	0.8142	0.7164	0.9431	17.6	-0.0074
+MSD	0.8108	0.8219	0.8228	0.7285	0.9558	17.8	0.0037
+RDrop	0.8122	0.8229	0.8242	0.7302	0.9577	23.1	0.0051
+EMA	0.8114	0.8235	0.8251	0.7312	0.9578	18.8	0.0043
ALL (full)	0.8043	0.8159	0.8162	0.7192	0.9502	24.5	-0.0028

Taxonomy ablation — 5-class vs 9-class, same recipe:

Taxonomy	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC	Minutes
5-class(ours)	0.8071	0.8194	0.8207	0.7249	0.9564	17.7
9-class(ours)	0.6096	0.8018	0.8076	0.7079	0.9456	17.6

Figure — Component ablation



Plain summary: Each bar shows what adding one component does on top of the CE+FGM reference. In this run, R-Drop is the strongest single add-on (+0.0051 Macro-F1), EMA and MSD also help slightly, while the balanced sampler and the full stack do not improve over the CE+FGM reference. This supports keeping **Model A** simple and treating **Model B** as a separate full-stack contrast rather than the single headline system.

Methodology note (honest). A *pre-flight* sanity check on a small **balanced** sample suggested only FGM helped; we flagged that a balanced sample structurally hides the benefit of the imbalance components, so the **real, imbalanced** NB08 numbers above are the authoritative ones that decide the final recipe. A surprising earlier finding — that **uniform LR beats LR decay** — is why the final config uses uniform learning rates.

14. Main Results — Benchmark (20% in-domain test)

5-class benchmark scheme (none/abusive/sexual/religious/threat).

System	Macro-F1	Weighted-F1	Accuracy	MCC	Macro-AUROC
Best baseline (TFIDF(word+char)+LinearSVM)	0.7674	0.7889	0.7933	0.6788	0.9418
Model A — Script-Aware Ensemble	0.8225	0.8332	0.8339	0.7452	0.9626
Model B — Full-Stack BanglishBERT	0.8135	0.8224	0.8222	0.7291	0.9534

Per-encoder single-model scores for Model A, from `outputs/models_main/per_run_summary.csv` :

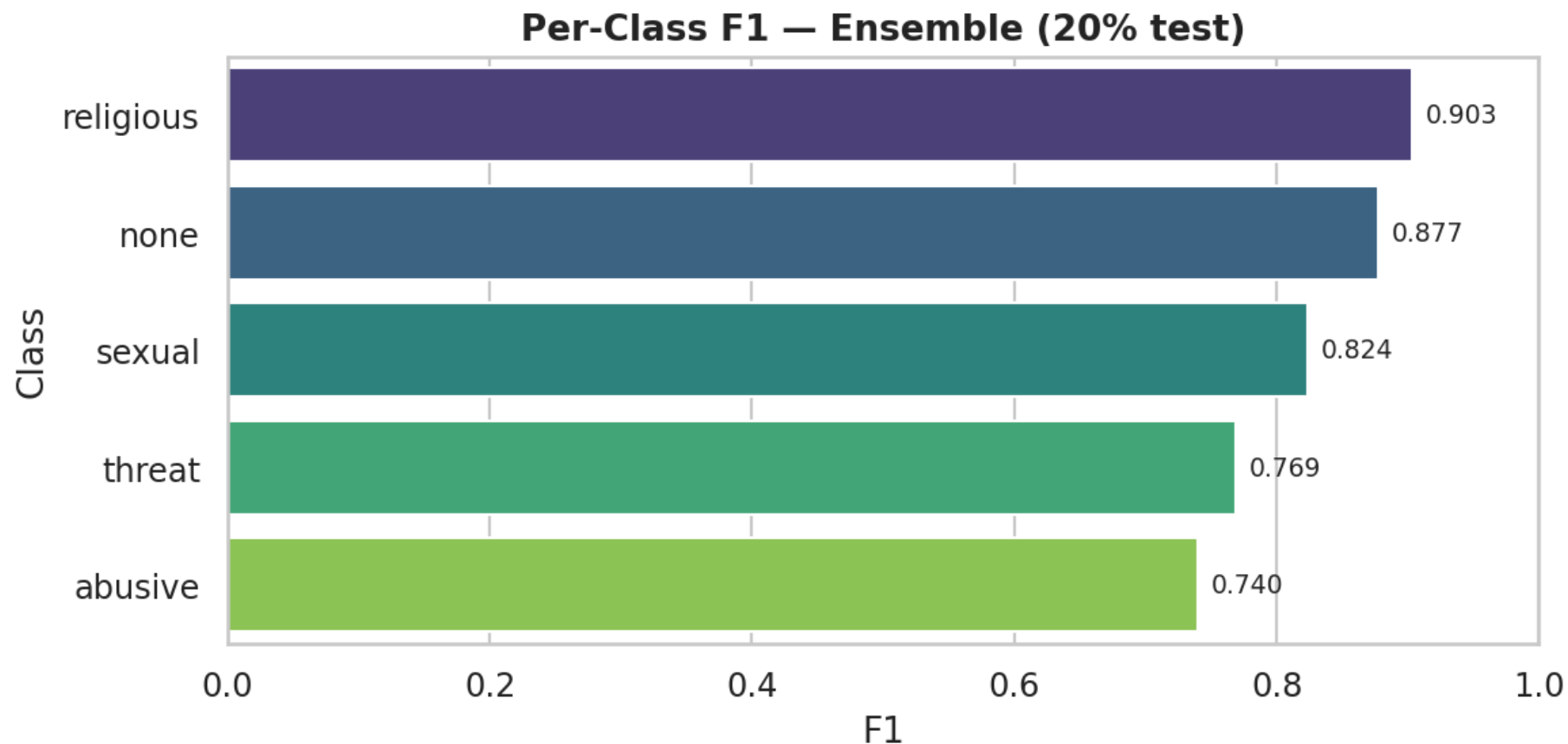
Model	Seed	Macro-F1	Weighted-F1	Accuracy	MCC	AUROC
banglishbert	42	0.8069	0.8201	0.8206	0.7252	0.9567
banglishbert	123	0.7996	0.8124	0.8161	0.7151	0.9547
banglishbert	456	0.8083	0.8200	0.8213	0.7261	0.9568
banglabert	42	0.8105	0.8178	0.8177	0.7545	0.9602
banglabert	123	0.8154	0.8211	0.8218	0.7603	0.9599
banglabert	456	0.8149	0.8205	0.8216	0.7600	0.9615
muril	42	0.8055	0.8177	0.8185	0.7231	0.9521
muril	123	0.8084	0.8216	0.8222	0.7284	0.9502
muril	456	0.8066	0.8198	0.8210	0.7259	0.9523
xlmr	42	0.8052	0.8159	0.8161	0.7199	0.9538
xlmr	123	0.8016	0.8144	0.8145	0.7166	0.9503
xlmr	456	0.8035	0.8173	0.8181	0.7215	0.9501

Model B per-seed validation Macro-F1: 0.8034 / 0.8018 / 0.8056; **seed-ensemble** lifts test Macro-F1 to 0.8135 (best single seed 0.8086).

14.1 Per-class F1 — final systems

Class	Model A F1	Model B F1
abusive	0.7397	0.7214
none	0.8771	0.8675
religious	0.9031	0.8953
sexual	0.8238	0.8226
threat	0.7689	0.7608

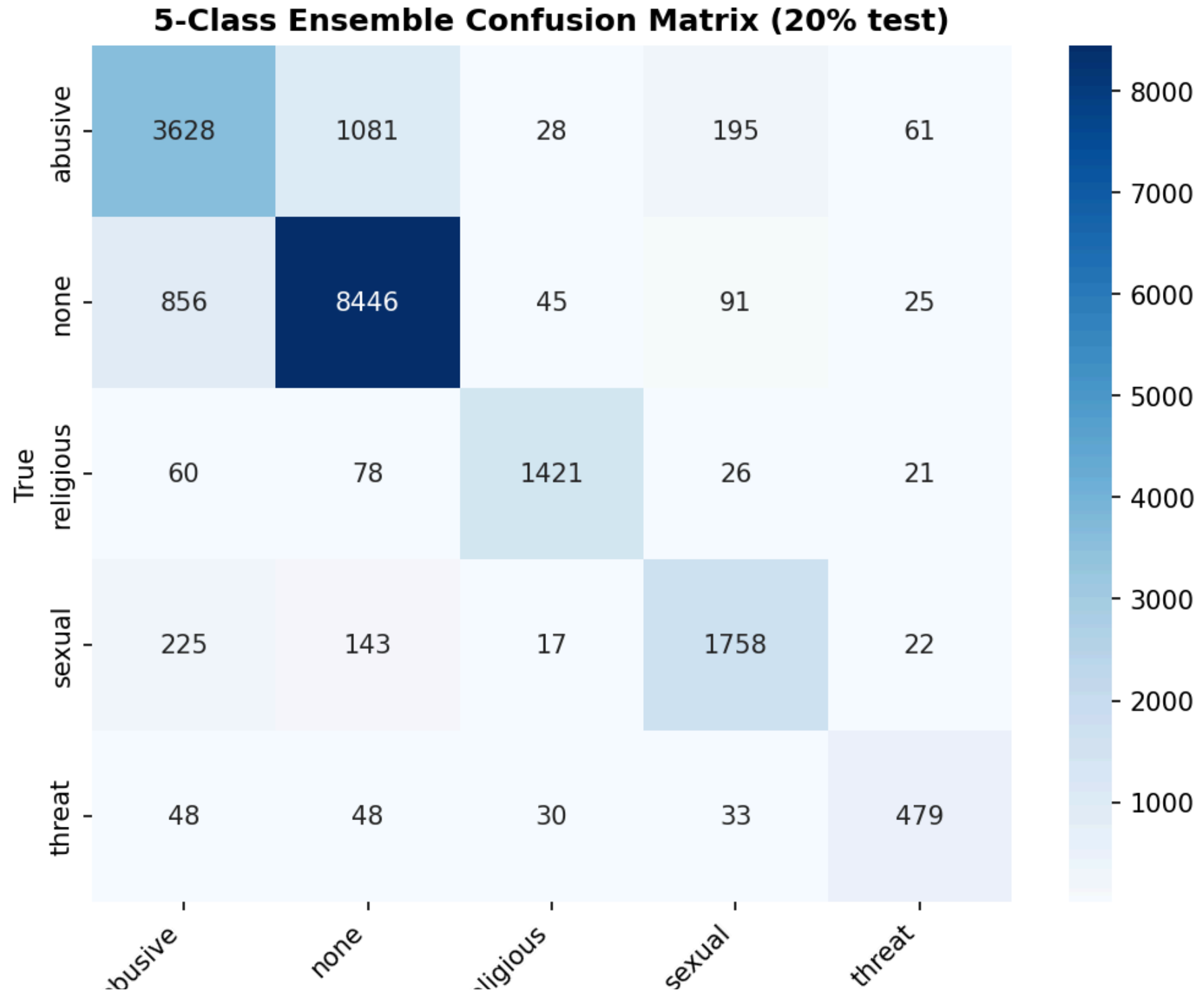
Figure — Per-class F1



Plain summary: `religious` and `none` are the strongest; `abusive` is the hardest in our scheme because it is the catch-all bucket that absorbs many heterogeneous behaviours (troll, personal, political, spam), making its boundary fuzzier than the semantically tight classes. Model A is the higher in-domain system overall, and it also improves every listed per-class F1 over Model B in the uploaded benchmark outputs.

14.2 Confusion matrix — Model A ensemble, 20% test

Figure — Confusion matrix (ensemble, 20% test)



Predicted

Numeric matrix, read from the final confusion-matrix output (`true` rows × `predicted` columns):

True \ Pred	abusive	none	religious	sexual	threat
abusive	3,628	1,081	28	195	61
none	856	8,446	45	91	25
religious	60	78	1,421	26	21
sexual	225	143	17	1,758	22
threat	48	48	30	33	479

Plain summary: Most mass sits on the diagonal. The off-diagonal mass concentrates in the `abusive ↔ none` region, especially true `abusive` predicted as `none` (1,081) and true `none` predicted as `abusive` (856). This is the same semantic-overlap difficulty the base paper reports between Troll and Not-Bully. `threat` remains the smallest class, but the model still recovers 479 true `threat` examples on the diagonal.

15. Robustness — the central study

15.1 Model B — Full-Stack BanglishBERT held-out evaluation

Model B held-out evaluation uses one seed per shifted config, while the in-domain benchmark uses the 3-seed ensemble.

Split	Held-out	n_test	Macro-F1	Weighted-F1	Accuracy	MCC	Macro-AUROC
script_holdout_bangla	bangla	56,989	0.1631	0.2617	0.3898	0.1088	0.6710
script_holdout_romanized	romanized	37,334	0.2165	0.6051	0.6425	0.0845	—
source_holdout_banth	banth	37,334	0.2165	0.6051	0.6425	0.0845	—

Split	Held-out	n_test	Macro-F1	Weighted-F1	Accuracy	MCC	Macro-AUROC
source_holdout_bd_shs	bd_shs	5,029	0.4549	0.5657	0.5383	0.3943	0.8153
source_holdout_facebook_44001	facebook_44001	43,078	0.5828	0.6256	0.6344	0.5186	0.8466
source_holdout_multilabel_12557	multilabel_12557	8,882	0.5579	0.5640	0.5907	0.4444	0.8524

15.2 Model A — Script-Aware Ensemble held-out evaluation

Model A robustness results are filled from the uploaded `robustness_summary(1).csv` output.

Split	Held-out	n_test	Macro-F1	Weighted-F1	Accuracy	MCC	Macro-AUROC
script_holdout_bangla	bangla	56,989	0.2218	0.3490	0.4618	0.2342	0.7162
script_holdout_romanized	romanized	37,334	0.2761	0.6833	0.6784	0.2278	—
source_holdout_banth	banth	37,334	0.2645	0.6915	0.6965	0.2390	—
source_holdout_bd_shs	bd_shs	5,029	0.4603	0.5814	0.5540	0.4110	0.8225
source_holdout_facebook_44001	facebook_44001	43,078	0.6037	0.6473	0.6572	0.5491	0.8811
source_holdout_multilabel_12557	multilabel_12557	8,882	0.5576	0.5577	0.5894	0.4506	0.8694

15.3 Robustness comparison: Model A vs Model B

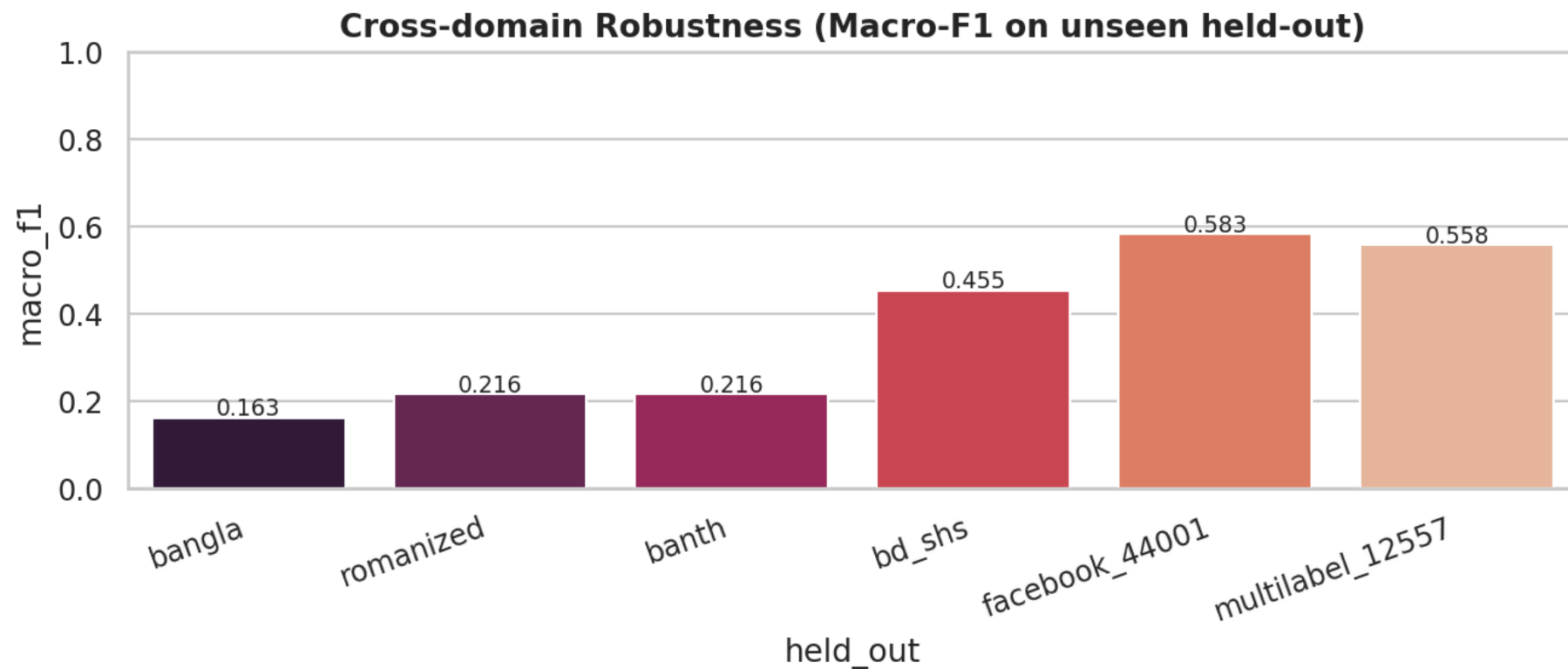
Split	Model B Macro-F1	Model A Macro-F1	A – B
script_holdout_bangla	0.1631	0.2218	0.0587
script_holdout_romanized	0.2165	0.2761	0.0596
source_holdout_banth	0.2165	0.2645	0.0480
source_holdout_bd_shs	0.4549	0.4603	0.0054

Split	Model B Macro-F1	Model A Macro-F1	A – B
source_holdout_facebook_44001	0.5828	0.6037	0.0209
source_holdout_multilabel_12557	0.5579	0.5576	-0.0003

Key observations.

1. **In-domain is strong; transfer degrades by type of shift.** Source shift (same script) costs a lot but stays usable, while **script shift remains much harder**.
2. **Cross-script transfer is the open wall.** Accuracy and Weighted-F1 can look acceptable under script shift because the model defaults toward the majority class, while Macro-F1 exposes weak minority-class transfer.
3. In the uploaded shifted-evaluation outputs, **Model A improves over Model B on most held-out settings**, including `script_holdout_bangla`, `script_holdout_romanized`, `source_holdout_bd_shs`, and `source_holdout_facebook_44001`. The `source_holdout_multilabel_12557` result is effectively tied, with Model B higher by only 0.0003 Macro-F1.
4. `source_holdout_banth` and `script_holdout_romanized` are conceptually linked because `banth` is the only Romanized source. For Model B the two rows are identical; for Model A the uploaded records are close but not identical, so they should be described as related, not as independent evidence.
5. This is exactly why **Model A is script-aware**: a single model cannot reliably bridge the scripts, so we split the work between a Bangla specialist (BanglaBERT) and bilingual/multilingual encoders.

Figure — Robustness across held-out splits



Plain summary: The drop from in-domain to script-held-out is the paper's most important empirical message: dual-script generalisation, not in-domain accuracy, is the real frontier for Bengali cyberbullying detection. Model A improves robustness but still leaves cross-script transfer as an open research problem.

16. Base-Paper Comparison (Facebook-44K, native 5-class)

Protocol matched to the base paper: Facebook 44,001 → within-source dedup **43,112** → native classes (Not Bully, Sexual, Troll, Religious, Threat) → **70/15/15** (train 30,178 / val 6,467 / test 6,467). Both proposed systems are retrained on this split.

Class / Metric	Base paper (H&S 2025)	Model B (ours)	Model A (ours)
Not Bully	0.9151	0.8813	0.8891

Class / Metric	Base paper (H&S 2025)	Model B (ours)	Model A (ours)
Sexual	0.8845	0.8711	0.8720
Troll	0.8446	0.8122	0.8192
Religious	0.9374	0.9366	0.9302
Threat	0.7579	0.8337	0.8292
Macro-F1	0.8923	0.8670	0.8679
Weighted-F1	—	0.8703	0.8736
Accuracy	—	0.8703	0.8737
MCC	—	0.8266	0.8314
Macro-AUROC	—	0.9719	0.9747

Model B: Weighted-F1 0.8703, Accuracy 0.8703, MCC 0.8266, Macro-AUROC 0.9719; per-seed val Macro-F1 0.8496 / 0.8508 / 0.8512.

Model A: Weighted-F1 0.8736, Accuracy 0.8737, MCC 0.8314, Macro-AUROC 0.9747.

Model A base-paper comparison weights:

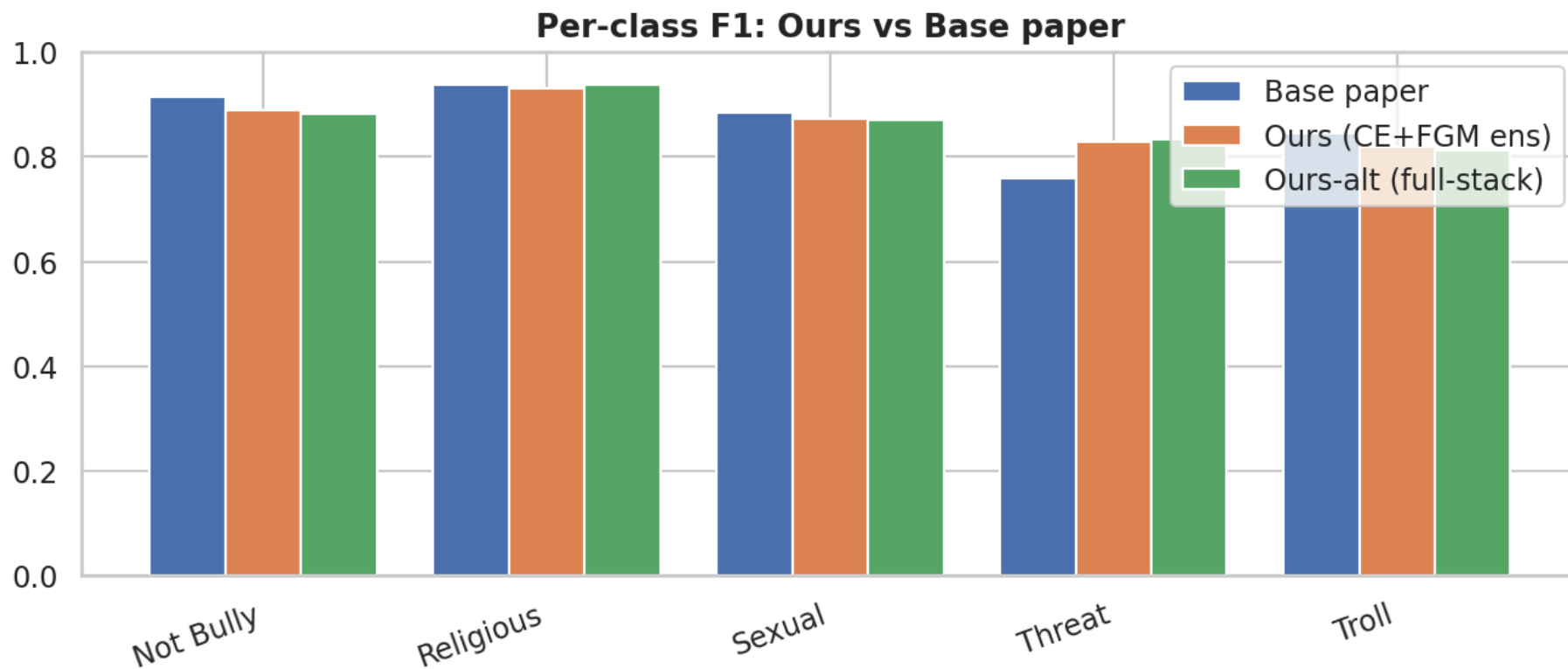
Run	Weight
banglishbert42	0.459955
banglabert42	0.076349
muril42	0.279786
xlmr42	0.183910

Model B base-paper seed-ensemble weights:

Seed	Weight
s42	0.501860

Seed	Weight
s123	0.370639
s456	0.127501

Figure — Base-paper comparison



Plain summary: On the base paper's own dataset our general-purpose systems are a touch below on overall Macro-F1 (Model A 0.8679, Model B 0.8670 vs base 0.8923) but **both beat it on the hardest class, Threat** — Model A by 0.0713 and Model B by 0.0758. We frame this honestly: the base paper is tuned to one clean source; our value is breadth (four sources, two scripts) and robustness, not a single-dataset win.

17. Error Analysis

- Macro vs. accuracy gap under script shift = **majority-class collapse** (Section 15).
- In-domain confusions concentrate on the `abusive` catch-all, mainly `abusive → none` and `none → abusive`.
- In the Model A confusion matrix, the largest off-diagonal cells are: true `abusive` predicted `none` = **1,081**, true `none` predicted `abusive` = **856**, true `sexual` predicted `abusive` = **225**, and true `abusive` predicted `sexual` = **195**.
- Hardest sources: `bd_shs` (small, different collection context) and `multilabel_12557` (multi-label annotation converted to single-label introduces label noise on transfer).
- Cross-script transfer remains the clearest failure mode: Model A improves shifted Macro-F1 over Model B but remains far below in-domain performance.
- Full per-source / per-script error breakdown is now represented by the two robustness tables in Section 15 and the confusion-matrix count table in Section 14.2.

18. Key Design Decisions

Decision	Rationale
Deduplicate (vs. keep)	honest held-out evaluation; no duplicate leakage
5-class (headline), 9-class (ablation only)	5 classes are well-populated and comparable to the base paper; 9 are sparse and drops Macro-F1 from 0.8071 to 0.6096
Priority <code>threat > sexual > religious > abusive > none</code>	keep explicit-violence and identity-targeted abuse from being absorbed into the generic bucket
Two proposed systems	RQ2: minimal script-aware committee vs. heavy single encoder
Script-isolated BanglaBERT	robustness shows cross-script transfer fails; specialise instead
CE + FGM for Model A	ablation-driven minimalism; CE+FGM is strong, and ensemble fusion lifts it to the best in-domain result
Uniform LR (no decay)	ablation showed decay hurt

Decision	Rationale
Macro-F1 primary	equal weight to rare classes (threat , religious)
3 seeds + ensemble	stability + a genuine ensemble signal

19. Limitations & Honest Claims

1. **Single-task, 5-class.** The system predicts a 5-class abuse type; do not claim severity prediction or a separate binary head as a headline (binary-equivalent = none F1 / non-none grouping).
2. `source_holdout_banth` and `script_holdout_romanized` are closely tied because `banth` is the only Romanized source; report them transparently and avoid counting them as fully independent evidence.
3. On the base paper's single clean dataset we are **slightly below** on overall Macro-F1; our claim is breadth + robustness + the Threat-class gain, not a single-dataset SOTA.
4. Robustness for Model B uses **1 seed per held-out config** (tractability); the in-domain and base-paper numbers use the full 3-seed ensemble.
5. The 0.5/0.5 CLS+mean pooling is a balanced empirical choice, not a proven optimum.
6. Model A improves shifted robustness in the uploaded outputs, but cross-script performance remains much lower than in-domain performance; do not overclaim this as solved.
7. The full-stack recipe does not automatically dominate: in the uploaded ablation, `ALL (full)` is below CE+FGM, while single add-ons such as R-Drop, EMA, and MSD show small gains.

20. Viva / Seminar Q&A

Q1. What is genuinely new here? A deduplicated four-source, dual-script Bengali benchmark; two contrasting proposed systems; and the first source-/script-held-out robustness study for this task.

Q2. Why two models instead of one? To answer whether a *minimal, script-aware committee* (A) can rival a *heavily regularised single encoder* (B). They differ on architecture and recipe at once, which is the comparison the paper is built around.

Q3. Which model is the headline now? Model A. It gives the best in-domain 20% test result: Macro-F1 **0.8225**, Weighted-F1 **0.8332**, Accuracy **0.8339**. Model B remains important as the full-stack single-encoder contrast.

Q4. Why is BanglaBERT restricted to Bangla? Because robustness shows cross-script transfer fails; feeding a Bangla-pretrained model Romanized text degrades it and pollutes the ensemble, so we mask it off Romanized rows.

Q5. Why does cross-script Macro-F1 crater while accuracy stays moderate? Majority-class collapse: on an unfamiliar script the model predicts `none` for many inputs, which is often correct (moderate accuracy) but kills minority-class recall (low Macro-F1).

Q6. You're below the base paper on Facebook — isn't that bad? On *overall* Macro-F1, marginally; but we **improve the hardest class** (`Threat`) and, unlike the base paper, generalise across four sources and two scripts. Benchmarks are judged on rigour and new axes, not one dataset's leaderboard.

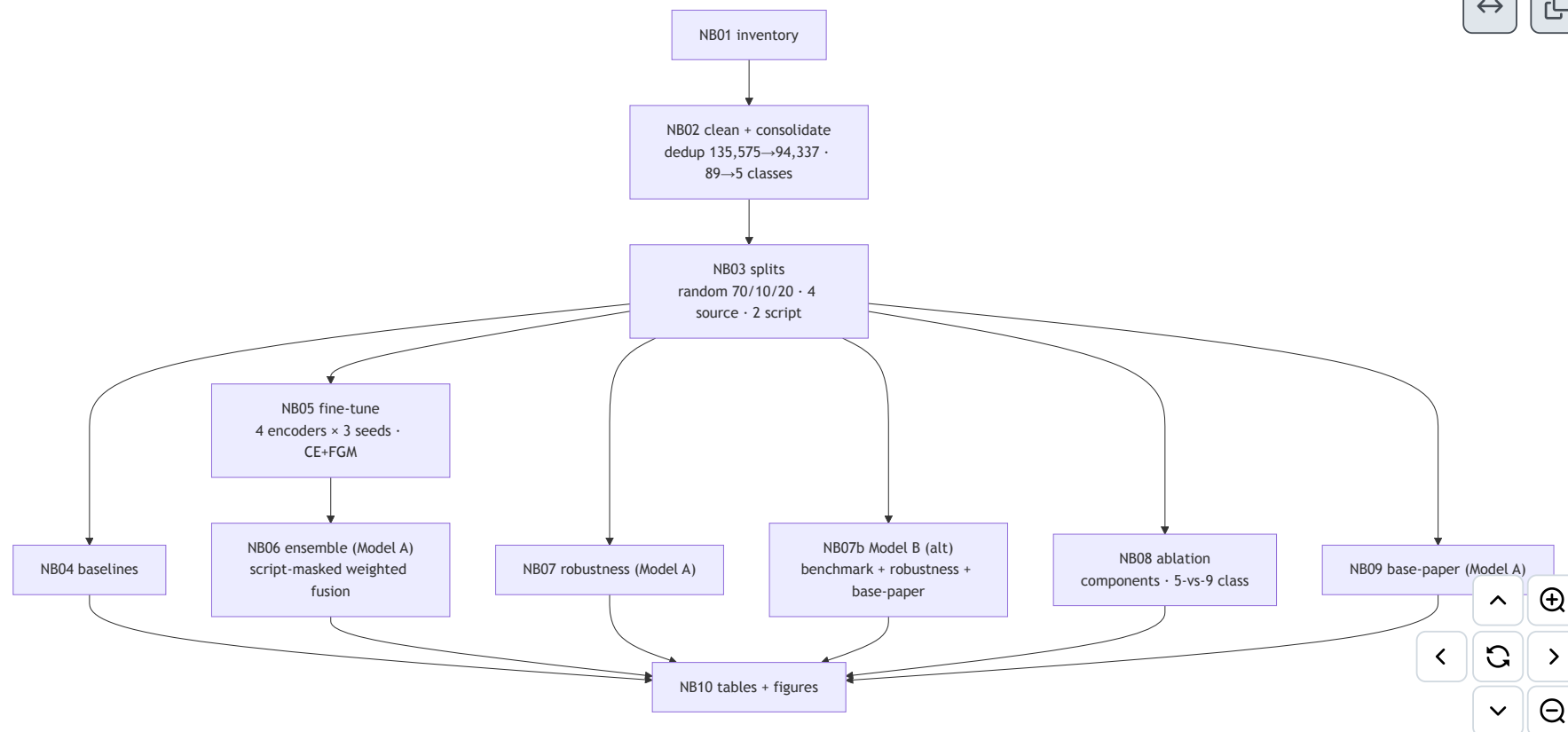
Q7. Why Macro-F1? It weights every class equally, so the rare `threat` / `religious` classes are not hidden by the dominant `none` class.

Q8. Why uniform LR? The ablation showed LR decay reduced performance on this multi-source, dual-script mixture.

Q9. Why deduplicate now? Duplicates (especially in `banth`) would leak across held-out splits and inflate robustness numbers; removing them makes the evaluation trustworthy.

Q10. What is the strongest baseline? TF-IDF word+char LinearSVM with Macro-F1 **0.7674**. It is strong but still behind Model A by **0.0551** Macro-F1.

21. Reproducibility — Pipeline & Outputs



Output tree (`outputs/` , per `output_dir.txt`):

```

outputs/
├─ baselines/baseline_results.csv
├─ models_main/           # NB05: 4 enc × 3 seeds (results.json), per_run_summary.csv
├─ ensemble/              # NB06 (Model A): ensemble_test_metrics.json, test_pred/proba.npy,
cm_ensemble_20pct.png
├─ ablation/              # NB08: component_ablation.csv, taxonomy_ablation.csv
├─ robustness/            # NB07 (Model A): robustness_summary.csv + per-config
├─ altmethod/             # NB07b (Model B): benchmark / robust_* / basepaper / *_summary.json
├─ basepaper/comparison.json # NB09 (Model A) vs base paper
└─ fig_label5_distribution.png
  
```

```
└─ paper/          # NB10: table1..6 (.csv/.tex), fig_per_class/ablation/robustness/basepaper.png
└─ paper/          # NB10: final tables and curated figures used in this log
```

Uploaded result files inserted into this updated log:

Uploaded file	Used for
baseline_results.csv	NB04 baseline table
component_ablation.csv	NB08 component ablation table
taxonomy_ablation.csv	NB08 5-vs-9-class taxonomy ablation
ensemble_test_metrics.json	Model A in-domain benchmark, per-class F1, weights, binary-equivalent metrics
per_run_summary.csv	Model A per-encoder/per-seed single-run results
robustness_summary(1).csv	Model A held-out robustness table
altmethod_summary.json	Model B benchmark, robustness, base-paper run
robustness_summary.csv	Model B held-out robustness CSV mirror
comparison.json	Model A base-paper comparison
basepaper_comparison.json	Model B base-paper comparison
label_encoder.json	final class → id mapping
cm_ensemble_20pct.png	Model A 20% test confusion-matrix figure
dataset_inventory_summary.csv	raw source-file inventory
dataset_folder_summary.csv	folder-level inventory summary
dataset_file_tree.csv	raw data file tree

22. Asset Index (figures used above)

#	Figure	Source file (<code>outputs/...</code>)	Status
01	Class distribution	<code>outputs/paper/fig_class_distribution.png</code>	exists
02	Confusion matrix (ensemble, 20%)	<code>outputs/paper/fig_confusion_matrix.png</code>	exists; numeric table inserted in Section 14.2
03	Per-class F1	<code>outputs/paper/fig_per_class_f1.png</code>	exists
04	Component ablation	<code>outputs/paper/fig_ablation.png</code>	exists; values inserted in Section 13
05	Robustness	<code>outputs/paper/fig_robustness.png</code>	exists; Model A and B tables inserted in Section 15
06	Base-paper comparison	<code>outputs/paper/fig_basepaper.png</code>	exists; Model A and B values inserted in Section 16

Figures are embedded from repo-relative `outputs/paper/fig_*.png` paths. If your curated filenames differ, adjust the six URLs (or point them at the source paths in the table above). All **diagrams** in this log are Mermaid placeholders to be replaced with rendered PNGs later.

23. Changelog vs. the previous draft

Previous draft	This version (current project)
Multi-task (binary + 9-class)	Single-task 5-class (9-class only in ablation)
3 encoders	4 encoders (adds BanglishBERT; BanglaBERT script-isolated)
Kept duplicates	Deduplicated (135,575 → 94,337)
80/10/10	70/10/20
One system	Two proposed systems (A: script-aware CE+FGM ensemble; B: full-stack BanglishBERT)
No script-aware masking	Script-aware ensemble (BanglaBERT masked on Romanized)

Previous draft	This version (current project)
Heavier default recipe	Model A is ablation-driven minimal (CE+FGM) ; Model B keeps the full stack
Placeholder metrics	All uploaded CSV/JSON metrics inserted
Model B was the only fully filled headline	Model A is now the main/best in-domain system; Model B remains the alternate full-stack contrast

24. Final Claim Boundary

Use the following claims in the paper and presentation:

1. **Safe headline claim:** BanglaCyberBench provides a deduplicated, multi-source, dual-script, 5-class Bengali cyberbullying benchmark with systematic source/script-held-out testing.
2. **Safe model claim:** the script-aware ensemble (Model A) is the best in-domain system in the final uploaded outputs, reaching Macro-F1 **0.8225** on the 20% test.
3. **Safe comparison claim:** on the base paper's own Facebook protocol, both proposed systems are slightly lower than H&S 2025 on overall Macro-F1 but improve the **Threat** class substantially.
4. **Safe robustness claim:** cross-script transfer remains the main unsolved challenge; Model A improves held-out robustness in most uploaded shifted settings but does not eliminate the gap.
5. **Avoid this claim:** do not say the method is new SOTA on the base paper's single-source Facebook dataset, because the base paper's Macro-F1 is still higher there.

BanglaCyberBench — Experiment Log v5 | June 2026 | Sefayet Alam, Naim Parvez and A. F. M. Minhazur Rahman